

Weihan Chen

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Education

Georgia Institute of Technology

B.S. in Computational Media | GPA: 3.83/4.00

Expected Dec 2027

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Computer Organization, Discrete Mathematics, Linear Algebra, Probability & Statistics

University of Utah

Previous Undergraduate Study in Games | GPA: 4.00/4.00

Projects

Project Rivering | C++ | Unreal Engine 5

Apr 2026 – Present

Independent UE5 action game with custom combat, movement, and procedural world systems

Combat & Melee Interaction

- Design C++ combat state and action arbitration across attacks, blocking, and targeting, integrating Unreal Gameplay Ability System (GAS) attributes and effects to enforce shared stamina constraints.
- Implement animation-timed melee collision sweeps between weapon poses, validate contact data before resolving hits, and track accepted targets per attack to prevent repeated hit processing across frames.

Character Movement & Networked Gameplay

- Integrate Unreal Mover layered movement for dashes and hit reactions, converting animation root motion into character displacement and handling blocking collisions within the movement update.
- Route Enhanced Input combat commands through server-side handlers and replicate action and hit-reaction state, connecting local controls to shared gameplay decisions and animation presentation.

Procedural Environment Tooling

- Build modular building assets with Python/Blender and integrate them into Unreal's Procedural Content Generation (PCG) workflow, exposing controls for building height, dimensions, and appearance.

Camera & Occlusion Handling

- Control wall and roof visibility through camera-to-character geometry queries and material clipping, preserving original collision geometry so visibility changes do not alter obstruction checks.

Friday Night | Python | FastAPI | SQLAlchemy

Jun 2026

Web application for collaborative movie discovery, selection, and planning

- Built a FastAPI backend with SQLAlchemy/SQLite for users, groups, invitations, movie selection, voting, and persistent shared state.
- Implemented signed-cookie authentication and role-based group authorization, including cross-group access controls, secure cookie settings, password hashing, and CSRF protection.
- Integrated the TMDB API and deployed the application behind Nginx/Uvicorn over HTTPS with environment-managed secrets and security-focused tests.

WayJia | GML | GameMaker Studio | GLSL ES

Sep 2025 – Dec 2025

Three-person experimental space-exploration game — sole programmer and producer

- Implemented physics-based spacecraft controls, radar exploration, environmental interactions, and level progression, integrating dialogue, audio, and cutscenes into a complete game flow.
- Ported and adapted Shadertoy effects to GameMaker GLSL ES, implementing runtime parameter binding, camera-aligned rendering, and off-screen render surfaces.

Technical Skills

Languages: C++, C, Java, Python, SQL

Frameworks / Engines: Unreal Engine 5, Django, FastAPI, SQLAlchemy, Gameplay Ability System (GAS)

Tools / Technologies: Git, SQLite, Blender, Unreal Blueprint, Enhanced Input